FEATURE 02 OF 14 | PRIORITY P0

Rule-Based Detection

Priority	P0
Judging criteria served	Innovation (20) | Completeness (35)
Estimated effort	Medium (~3-5 hrs) (rough estimate)
Depends on	Message Input

Purpose

This is the engine. It reads the message and finds specific, named warning signs using visible rules - not a black-box model. Because every rule is written down and shows its evidence, users and judges can see exactly why a message was flagged. That transparency is the main innovation claim.

What it must do

- Detect five signal types: **urgency**, **account threats**, **suspicious links**, **OTP requests**, **password / PIN / card-detail requests**.
- Each rule returns: rule id, human label, category, points, and the exact matched text with start/end positions.
- Rules are stored as data (a JSON/JS config object), not hard-coded in logic, so weights and phrases can be edited without touching code.
- A rule fires once for scoring even if it matches many times (see feature 3), but every match is still returned for highlighting.
- The engine is a pure function: analyse(text, config) returns a result object. No hidden state, so it is easy to unit-test.

How it should work

- **Starter rule table** (points are the values from your brief; the password rule is my proposal and is configurable):
- URGENCY +20 - phrases such as "immediately", "urgent", "within 24 hours", "act now", "expires today", "last chance".
- ACCOUNT_THREAT +15 - "account will be blocked / suspended / closed", "KYC expired / pending", "SIM will be deactivated", "legal action".
- SUSPICIOUS_LINK +25 - a link that has a risk marker (see feature 14): URL shortener, IP address instead of a name, odd ending such as .xyz/.top, punycode (xn--), an @ sign in the URL, or a bank name inside an unofficial domain. A plain link on its own is NOT suspicious.
- OTP_REQUEST +40 - the message asks the reader to share, send, forward or enter an OTP/verification code.
- PASSWORD_REQUEST +40 - asks for password, PIN, CVV, card number or login details. Shares a 'credential' group with OTP so it is not double-counted.
- **Negation handling (important):** genuine bank messages say "Never share your OTP". Before counting a match, look at roughly the 40 characters before it; if it contains "do not", "never", "don't", "will never ask" the match is treated as a safety advisory and ignored.
- **Verb + object pattern for credentials:** match a sharing verb (share, send, tell, reply with, enter, provide) near the word OTP/PIN/password. The word OTP alone is not a warning sign - banks legitimately send OTPs.
- Matching uses case-insensitive regular expressions with word boundaries; keep each rule's patterns in a list so new phrases are one-line additions.

UI / UX notes

- The user never sees raw regular expressions - only plain labels ("Fake urgency", "Asks for your OTP").
- Show a small "5 checks run" line so a clean result still shows work was done.

Edge cases and risks

- "Your OTP is 482913. Do not share it with anyone." - a real OTP delivery message. Must score Low (negation rule + no request verb).
- "Please share your feedback within 24 hours" - contains urgency wording but no threat. Urgency alone is only +20, so stays Low.
- Misspellings and spacing tricks ("O T P", "0TP", "urgnt"): add a few common variants; do not try to catch everything in the MVP.
- Same phrase repeated five times: counts once for points.

Acceptance tests

Try this	Expected result
"Share your OTP immediately or account will be blocked"	Fires OTP_REQUEST, URGENCY, ACCOUNT_THREAT
"Never share your OTP with anyone"	No OTP_REQUEST (negation)
"Visit https://sbi.co.in for details"	No SUSPICIOUS_LINK (official domain on allowlist)
"Click bit.ly/x9A to update KYC"	SUSPICIOUS_LINK fires (shortener)
"Lunch at 1pm?"	Nothing fires; result Low / 0

Demo tip for the judges

> Open the rules table in the app (an expandable 'How detection works' section). Judges see real, readable rules - the opposite of a black box.